Daily Practice 3

Topics: Limits, Continuity, and Derivatives (Rules & Applications)

Section 1: Multiple Choice Questions

No calculator is allowed for these questions.

1. What is the value of $\lim_{x\to 3} \frac{x^2-2x-3}{x^2-9}$?

(A) 0

(B) $\frac{2}{3}$

(C) 1

(D) The limit does not exist.

2. What is the value of $\lim_{x\to 0} \frac{\sqrt{x+25}-5}{x}$?

(A) 0

(B) $\frac{1}{10}$

(C) $\frac{1}{5}$

(D) The limit does not exist.

3. Evaluate the limit: $\lim_{x\to 0} \frac{\sin(5x)}{2x}$

(A) 0

(B) $\frac{2}{5}$

(C) $\frac{5}{2}$

(D) 1

4. What is the value of $\lim_{x\to 0} \frac{1-\cos(3x)}{x}$?

(A) 0

(B) 3

(C) $\frac{3}{2}$

(D) The limit does not exist.

5. Let $f(x) = \frac{x^2-4}{|x-2|}$. What is $\lim_{x\to 2^-} f(x)$?

(A) -4

(B) -2

(C) 4

(D) The limit does not exist.

6. Which of the following lines is a horizontal asymptote to the graph of $y=\frac{4x^2-5x+2}{3x^2+7}$?

(A) $y=0$

(B) $y=\frac{4}{3}$

(C) $y=\frac{3}{4}$

(D) There is no horizontal asymptote.

7. What is the value of $\lim_{x\to\infty}\frac{2x^3-x^2+5}{5x^4+3x}$?

(A) 0

(B) $\frac{2}{5}$

(C) $\frac{5}{2}$

(D) ∞

8. What is the value of $\lim_{x\to-\infty}\frac{3x}{\sqrt{4x^2+1}}$?

(A) $-\frac{3}{2}$

(B) 0

(C) $\frac{3}{2}$

(D) The limit does not exist.

9. Let f be the piecewise function defined below:

$$f(x)=\begin{cases} x^2-ax, & \text{for } x<2 \\ 4-x, & \text{for } x\geq 2 \end{cases}$$

For what value of the constant a is f continuous at $x=2$?

(A) -1

(B) 0

(C) 1

(D) 2

10. The function g is continuous on the closed interval $[0, 5]$. If $g(0) = 2$ and $g(5) = -4$, which of the following statements MUST be true?

(A) $g(x) = 0$ for at least one x between 0 and 5.

(B) $g(c) = -1$ for some c in $(0, 5)$.

(C) The maximum value of $g(x)$ on $[0, 5]$ is 2.

(D) Both (A) and (B) must be true.

11. At which x-value does $f(x) = \frac{x^2-x-12}{x^2-16}$ have a removable discontinuity?

(A) $x = -4$

(B) $x = 4$

(C) $x = 3$

(D) $x = -3$

12. Which of the following limits represents the derivative of $f(x) = \sqrt{x}$ at $x = 4$?

(A) $\lim_{h \to 0} \frac{\sqrt{4+h}-2}{h}$

(B) $\lim_{x \to 4} \frac{\sqrt{x}-4}{x-4}$

(C) $\lim_{h \to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h}$

(D) Both (A) and (B) are correct.

13. Evaluate the limit: $\lim_{h \to 0} \frac{\sin\left(\frac{\pi}{6}+h\right)-\sin\left(\frac{\pi}{6}\right)}{h}$

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) 1

14. The graph of a function f has a sharp corner at the point $(2, 5)$. Which of the following statements is true?

(A) f is not continuous at $x = 2$.

(B) $f'(2)$ does not exist.

(C) $\lim_{x \to 2} f(x)$ does not exist.

(D) $f(2)$ is undefined.

15. If $y = \frac{3}{x^2} - 5\sqrt{x} + \pi^2$, what is $\frac{dy}{dx}$?

(A) $-\frac{6}{x^3} - \frac{5}{2\sqrt{x}}$

(B) $-\frac{6}{x^3} - \frac{5}{2\sqrt{x}} + 2\pi$

(C) $-\frac{3}{x^3} - \frac{5}{2\sqrt{x}}$

(D) $-\frac{6}{x} - \frac{5}{2}x^{-1/2}$

16. If $f(x) = x^3e^x$, then $f'(x) = $

(A) $3x^2e^x$

(B) $x^3e^x + 3x^2e^x$

(C) $x^3e^x - 3x^2e^x$

(D) $3x^2e^{x-1}$

17. If $g(x) = \frac{\sin x}{x^2}$, what is $g'(x)$?

(A) $\frac{\cos x}{2x}$

(B) $\frac{x^2 \cos x + 2x \sin x}{x^4}$

(C) $\frac{x^2 \cos x - 2x \sin x}{x^4}$

(D) $\frac{2x \sin x - x^2 \cos x}{x^4}$

18. What is the slope of the line tangent to the graph of $y = \ln(3x)$ at $x = \frac{1}{3}$?

(A) 1

(B) 3

(C) $\frac{1}{3}$

(D) 9

19. If $f(x) = \cos(x^3)$, then $f'(x) = $

(A) $-\sin(x^3)$

(B) $-3x^2 \sin(x^3)$

(C) $3x^2 \sin(x^3)$

(D) $-\sin(3x^2)$

20. Let $h(x) = (2x+1)^5(x^2-1)^3$. Which of the following is equivalent to $h'(x)$?

(A) $5(2x+1)^4(x^2-1)^3+3(2x+1)^5(x^2-1)^2$

(B) $10(2x+1)^4(x^2-1)^3+6x(2x+1)^5(x^2-1)^2$

(C) $10(2x+1)^4 \cdot 6x(x^2-1)^2$

(D) $30x(2x+1)^4(x^2-1)^2$

21. If $y=e^{\tan x}$, then $\frac{dy}{dx} = $

(A) $e^{\sec^2 x}$

(B) $e^{\tan x} \sec^2 x$

(C) $\ln(\tan x) \sec^2 x$

(D) $e^{\tan x} \cot x$

22. If $x^2+y^2=25$, what is the value of $\frac{dy}{dx}$ at the point $(3,4)$?

(A) $-\frac{3}{4}$

(B) $\frac{3}{4}$

(C) $-\frac{4}{3}$

(D) $\frac{4}{3}$

23. Consider the curve defined by $xy+y^2=6$. What is the slope of the tangent line at $(1,2)$?

(A) $-\frac{2}{5}$

(B) $-\frac{1}{3}$

(C) $\frac{1}{2}$

(D) $-\frac{4}{5}$

24. If $y=\arcsin(2x)$, what is $\frac{dy}{dx}$?

(A) $\frac{2}{\sqrt{1-4x^2}}$

(B) $\frac{1}{\sqrt{1-4x^2}}$

(C) $\frac{2}{\sqrt{1-x^2}}$

(D) $\frac{2}{1+4x^2}$

25. Let f and g be differentiable functions such that $g(x)=f^{-1}(x)$. If $f(2)=5$ and $f'(2)=4$, what is the value of $g'(5)$?

(A) 4

(B) $\frac{1}{4}$

(C) $-\frac{1}{4}$

(D) $\frac{1}{5}$

26. If $f(x) = x^4 - 2x^3 + 5x$, what is $f''(x)$?

(A) $4x^3 - 6x^2 + 5$

(B) $12x^2 - 12x$

(C) $12x^2 - 12x + 5$

(D) $24x - 12$

27. The position of a particle moving along a straight line at time t is given by $s(t) = t^3 - 6t^2 + 9t$. At what time(s) t is the particle at rest?

(A) $t = 3$ only

(B) $t = 1$ only

(C) $t = 1$ and $t = 3$

(D) The particle is never at rest.

28. Evaluate the limit: $\lim_{x\to\infty} x \sin\left(\frac{1}{x}\right)$

(A) 0

(B) 1

(C) ∞

(D) The limit does not exist.

29. Let $y = \frac{\ln x}{x}$. What is y' ?

(A) $\frac{1}{x^2}$

(B) $\frac{1-\ln x}{x^2}$

(C) $\frac{\ln x - 1}{x^2}$

(D) $\frac{1}{x}$

30. If $f(x) = 2^{3x}$, then \$f'(x) = \$

(A) $3 \cdot 2^{3x}$

(B) $2^{3x} \ln 2$

(C) $3 \cdot 2^{3x} \ln 2$

(D) $3x \cdot 2^{3x-1}$

Section 2: Free Response Questions

Show all your work clearly.

FRQ 1. Let f be the piecewise function defined by:

$$f(x) = \begin{cases} \frac{x^2-3x-10}{x-5}, & \text{for } x < 5 \\ k, & \text{for } x = 5 \\ ax^2 + b, & \text{for } x > 5 \end{cases}$$

where k, a, and b are constants.

(a) Evaluate $\lim_{x \to 5^-} f(x)$. Show the algebraic steps that lead to your answer.

(b) If f is continuous at $x = 5$, what must be the value of k? Justify your answer using the mathematical definition of continuity.

(c) The function f is differentiable at $x = 5$. Find the values of a and b. Show the work that leads to your answers.

FRQ 2. Consider the curve given by the equation $y^3 - xy = 2$.

(a) Find $\frac{dy}{dx}$ in terms of x and y.

(b) Write an equation for the line tangent to the curve at the point $(-1, 1)$.

(c) Find the coordinates of all points on the curve where the tangent line is vertical. Justify your answer.

(d) Evaluate $\frac{d^2y}{dx^2}$ at the point $(-1, 1)$.

FRQ 3. The table below gives selected values for two differentiable functions f and g, as well as their derivatives f' and g'.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	-2	-1	4
2	-1	5	1	-3
3	4	2	2	5

(a) Let $h(x) = f(x)g(x) - 2x$. Find $h'(1)$.

(b) Let $p(x) = \frac{f(x)}{g(x)+3}$. Find $p'(2)$.

(c) Let $m(x) = f(g(x))$. Write the equation of the line tangent to the graph of $y = m(x)$ at $x = 3$.

(d) Let $k(x) = \sqrt{f(x) + 6}$. Find $k'(3)$.

FRQ 4. A particle moves along the x-axis so that its position at time $t > 0$ is given by $x(t) = \frac{t^3}{3} - 3t^2 + 8t$.

(a) Find the velocity function $v(t)$ and the acceleration function $a(t)$.

(b) Find all times $t > 0$ when the particle is at rest.

(c) On the interval $0 < t < 5$, when is the particle moving to the left? Justify your answer.

(d) Is the speed of the particle increasing or decreasing at $t = 3$? Give a reason for your answer.